

RESEARCH ARTICLE

Quantum Kernel-Driven Hybrid Neural Network for Brain Tumor Classification

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ABSTRACT Multiclass brain tumor classification from magnetic resonance imaging (MRI) remains challenging due to subtle inter-class visual similarities, particularly for anatomically small and heterogeneous tumor types such as pituitary lesions. To address these challenges, this paper presents a hybrid classical-quantum learning framework that integrates a deep convolutional feature extractor from the ResNet family with a variational quantum neural network (QNN) head for enhanced feature representation. In the present work, classical deep features extracted by the ResNet model are encoded into a six-qubit quantum circuit, enabling richer feature interactions within a high-dimensional Hilbert space. In addition, an anatomically guided Kernel-based Center Cropping (KCC) strategy is introduced as a deterministic spatial prior to emphasize anatomically relevant central brain structures, particularly benefiting challenging pituitary tumor cases. The proposed Kernel-based Center Cropping-Hybrid Quantum Neural Network (KCC-HQNN) framework is trained and evaluated on a publicly available composite four-class brain MRI dataset curated from Kaggle sources, comprising glioma, meningioma, pituitary tumor, and no-tumor categories. Experimental results across multiple ResNet variants demonstrate that the hybrid quantum-classical models consistently outperform their purely classical counterparts, achieving classification accuracy of up to 99%, while macro-averaged precision, recall, and F1 scores consistently fall in the range of approximately 0.95–0.99 across different ResNet variants. Reliability and reproducibility are further validated through stratified five-fold cross-validation and repeated experiments using multiple random initialization seeds. This yields a mean classification accuracy of $99.00 \pm 0.08\%$, macro F1-score of 0.986 ± 0.11 , and weighted F1-score of 0.989 ± 0.12 , while cross-dataset evaluation on independent IEEE DataPort MRI datasets demonstrates robust generalization under dataset shift. All experiments are conducted on publicly available retrospective datasets using a quantum simulator. These results indicate that combining quantum feature processing with anatomically informed spatial processing improves multiclass discrimination of brain tumors.

INDEX TERMS Quantum machine learning, brain tumor, quantum transfer learning, convolutional neural network, kernel-based center cropping.

I. INTRODUCTION

Quantum computers are a novel category of computing machines, initially conceptualized by Richard Feynman in 1982 [1]. Inspired by Feynman's pioneering vision, in 1985, David Deutsch introduced the Quantum Turing Machine, which opened the possibility of taking classical computations

into the quantum realm via the superposition states of quantum bits (Deutsch, 1985). It is also worth noting that these early discoveries laid the foundation for what we now call quantum computing, which can solve some problems faster than classical methods [2]. Shor's algorithm, for instance, demonstrated quantum computers' potential by enabling polynomial-time factorization of integers, a task previously achievable only in exponential time on classical computers [3]. The Grover's algorithm also raised the complexity of

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database search problems to a new level, again proving the worth of quantum machines [4]. These innovations prompted remarkable technology corporations such as Google, Xanadu, Microsoft, IBM, and D-Wave to invest heavily in developing quantum technologies, many of which are now available via cloud computing platforms [5], [6], [7], [8]. With growing interest in cutting-edge technology, researchers have begun integrating quantum computing with machine learning, leading to the emergence of QML. This revolutionary field leverages the principles of quantum machines to enhance the capabilities of learning algorithms [9], [10], [11]. By harnessing quantum superposition, entanglement, and parallelism, Quantum Machine Learning (QML) addresses complex computational challenges more efficiently, pushing the boundaries of what traditional machine learning can achieve. This is particularly appropriate in fields like medical imaging, where the sheer complexity and dimensionality of data, such as MRI scans, pose significant computational challenges for classical models [12], [13], [14], [15].

Brain tumors are among the most complex and life-threatening disorders in medicine, exhibiting a high mortality rate. These tumors arise from abnormal cell proliferation, leading to uncontrolled growth that compromises healthy brain tissue. In this study, four types of brain tumors have been considered. Their distinct characteristics are visually represented in Fig. 1(a–d), providing a clear understanding of the differences among the selected tumor types as in MRI brain scans. They may originate from different cell types, including neurons, glial cells, or even the meninges, affecting the brain by increasing intracranial pressure and impairing vital neurological functions [16]. In the United States, it is estimated that around 18,760 individuals are diagnosed annually with malignant brain and central nervous system tumors [17], [18], [19].

In recent studies, researchers have explored deep learning approaches for brain tumor classification of MR images; notably, a hybrid deep learning approach, DeepTumorNet, achieved 99.6% accuracy in three-class classification [20]. Another approach of 3 attention-convolutional-LSTM (3ACL), a deep learning model leaning on 3D MRI, namely BRATS 2015 and BRATS 2018 datasets, with 98.90% and 99.29%, respectively, with only two classes of high-grade gliomas (HGG) and low-grade gliomas (LGG) [21]. Using Aquila Optimizer (AQO) along with VGG-16, VGG-19, and Inception-V3, a classification accuracy of 97.2%, 98.95%, and 97.38% was achieved for the brain tumor classification for two-class classifications [22]. Exploring the advantage of deep learning increases the accuracy of detection of brain tumors with various techniques such as U-Net [23], ResNet-18 [24], DensNet [25], GoogleNet [26], Xception [27], Ait et al. [22] achieved a validation accuracy of 98.70 by using Bayesian optimizer, whereas exploring the quantum machine learning hybrid approach resourcefully achieved this accuracy with an exponential reduction in cost. In this area, Dong et al. [28] proposed an HQN-CNN based

classification model achieving 97.8% accuracy across four classes, while Muhammad Asfand Hafeez et al. reached 90.1% accuracy in binary classification using a two-qubit setup [29]. To further explore the clinical potential of these architectures, recent studies have demonstrated the efficacy of quantum-classical hybrid systems for broader medical image classification tasks [53]. Quantum-inspired neural networks specifically for high-precision brain tumor localization [54]. While these models validate the growing interest in hybrid QML, they often struggle with high-dimensional feature handoff or small-lesion localization, which our proposed KCC-HQNN framework addresses through deterministic spatial constraints and an optimized 6-qubit variational circuit. Furthermore, advances in automated detection systems, such as developing deep neural networks and hybrid quantum computing models, are advancing medical diagnostics [30]. Transfer learning approaches have been employed to improve the accuracy and efficiency of tumor detection models by pre-training them on existing MRI datasets. Quantum-inspired feature processing offers enhanced representational expressivity rather than computational speedup in the current NISQ setting. This emerging technology provides unprecedented precision in recognizing complex patterns in medical images, allowing for more accurate diagnoses [31]. Recent research has proposed hybrid models combining classical neural networks with quantum-based methods to leverage the strengths of both technologies [28], [32], [33], [34]. Studies support the potential for quantum computers to transform medical imaging, particularly in brain tumor detection. Notable advancements in Noisy Intermediate-Scale Quantum (NISQ) technology [35] and fault-tolerant quantum systems underline this promising potential [36]. One of the most challenging aspects of brain tumor classification is the detection of pituitary gland tumors. These tumors are inherently difficult to identify due to their small size, proximity to surrounding structures, and subtle distinguishing features on MRI scans. Traditional methods often struggle to isolate the gland from overlapping tissue or to capture the intricate details needed for precise classification with 94.36 validation accuracy [37]; S. Gargya and S. Jain utilized CAD System Design for Pituitary Tumor Classification, achieving 96.00 accuracy [38].

II. QUANTUM COMPUTING

Represents a cutting-edge paradigm that leverages the principles of quantum mechanics to revolutionize computational capabilities. This shift enables quantum machines to process information in ways that classical systems cannot, providing unprecedented cryptography, optimization, and machine learning opportunities [39], [40], [41].

A. QUANTUM BITS (QUBITS)

In classical computing, the smallest data unit is a bit, i.e., 0 or 1. In contrast, quantum computing employs qubits, which can exist in a superposition of states, denoted as $|0\rangle$ and $|1\rangle$.

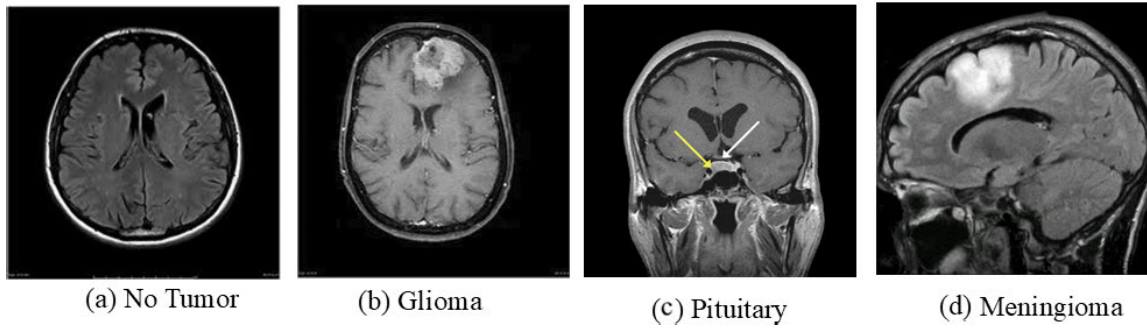


FIGURE 1. MR scans of the four considered categories.

B. QUANTUM GATES

Quantum gates manipulate qubits, analogous to classical logic gates, but with a fundamental difference: quantum gates operate on the probabilities and phases of qubits [42].

C. ENCODING CLASSICAL DATA TO A QUANTUM STATE

1) AMPLITUDE ENCODING

Amplitude encoding is a technique that encodes classical data values into the amplitudes of a quantum state. For a classical input vector $x = (x_1, \dots, x_N)$ of dimension $N = 2^n$, classical data are encoded into an n -qubit quantum state $|\psi(x)\rangle$ as:

$$U(x) : x \in \mathbb{R}^N \rightarrow \psi(x) = \frac{1}{\|x\|} \sum_{i=1}^n x_i i$$

where $\mathbb{R}^N$ denotes real numbers of N dimensions. In contrast, amplitude encoding can optimize the $O(\log(N))$ parameter, but in a quantum circuit, the depth of amplitude encoding increases up to $O(\text{poly}(N))$.

2) ANGULAR ENCODING

Angular encoding utilizes N qubits and maintains a constant quantum circuit depth. Qubit encoding rescales classical data x_i between 0 and π and inputs it into a single qubit as:

$$U(x) : x \in \mathbb{R}^N \rightarrow \psi(x) = \bigotimes_{i=1}^N \left(\cos\left(\frac{x_i}{2}\right) |0\rangle + \sin\left(\frac{x_i}{2}\right) |1\rangle \right),$$

for $i = 1 \dots N$

D. QUANTUM SIMULATION

Hamiltonian simulation, quantum annealing, and quantum lattice models address quantum physics, optimization, and phase estimation [2], [42], [43], [44]. Algorithms such as the variational quantum eigensolver (VQE) and quantum phase estimation (QPE) are used to compute molecular ground states and estimate Hamiltonian eigenvalues [45], [46], [47]. Due to decoherence, the error rate is high, and the problem is more complex, which is useful for addressing phenomena in drug discovery, materials science, and energy exploration that are inaccessible to classical methods. Circuit-based classifiers underperform when paired with classical preprocessing, limiting quantum benefit as well

as dimensional reduction [48]. By mapping features into quantum state spaces using unitary transformation, variational circuits exploit superposition to improve classification and computational efficiency [3]. For instance, IBMQX2 (5 qubits) outperforms IBMQ15-Melbourne (14 qubits) because it uses fewer Fredkin gates. Shor's algorithm breaks encryption by factoring integers exponentially faster [3], Grover's algorithm rapidly searches data quadratically [4], and quantum phase estimation aids quantum simulation while increasing sampling accuracy, proving quantum supremacy.

E. QUANTUM TRANSFER LEARNING

Mari et al. introduced a hybrid model intermingling classical transfer learning for feature extraction with a quantum circuit for image classification in identifying diabetic retinopathy [49]. Magallanes et al. proposed a hybrid transfer-learning framework for stenosis recognition by enhancing a pre-trained classical network with a quantum network [34]. Azevedo et al. explored a hybrid classical-quantum neural network using transfer learning for breast cancer detection [50]. Shahwar et al. developed a model for Alzheimer's disease detection employing ResNet34 for feature extraction and processing a 512-feature vector through a Quantum Variational Circuit (QVC) to generate precise decision boundaries [51]. Transformer-based models like Swin-UNETR have demonstrated remarkable performance in segmenting brain tumor MRI images, offering improved precision in challenging pituitary tumor cases [52]. As hybrid AI-quantum techniques evolve, they hold great promise for addressing these diagnostic complexities, ultimately advancing clinical outcomes.

F. RESEARCH GAP

Although deep convolutional neural networks (CNNs) have shown strong performance in brain MRI analysis, most existing methods rely on globally resized inputs and learn spatially diffuse representations without explicitly enforcing anatomical focus. This can reduce sensitivity to small and centrally located tumors such as pituitary lesions, where subtle tumor-specific cues are often overshadowed by surrounding dominant brain structures. In parallel, early quantum-assisted approaches for medical imaging have been explored only in limited or simplified settings and lack

systematic evaluation in realistic multi-class and cross-dataset scenarios. Importantly, there is no unified framework that simultaneously constrains the field of view toward clinically relevant central brain regions and leverages quantum feature encoding to enhance the separability of subtle inter-class variations. Consequently, the potential of hybrid quantum–classical learning for anatomy-aware, pituitary-sensitive tumor characterization remains underexplored. To address this gap, we propose an anatomy-guided hybrid quantum–classical framework that emphasizes central brain structures through deterministic center cropping and maps compact classical features into a quantum representation to better capture subtle, hard-to-discern tumor patterns, with comprehensive evaluation of performance, scalability, and generalization across diverse brain MRI datasets.

G. RESEARCH OBJECTIVE

Quantum-based models, such as Quantum Transfer Learning (QTL), have the potential to provide reliable and accurate classification of complex medical image data. Further, there is a need to evaluate and conduct comprehensive analyses of the application of quantum models, including performance, scalability, and their ability to generalize across different medical datasets. Accordingly, we have proposed an algorithm that uses a kernel-based center-cropping technique, specifically helpful for extracting features of the pituitary gland and other brain regions, in combination with the hybrid quantum-classical model (KCC-HQNN). The proposed algorithm enhances the ability to recognize subtle patterns specific to pituitary tumors, which might otherwise go unnoticed. As a result, overall MR brain image classification improves. The present work also evaluates the performance of the proposed model in terms of reproducibility and reliability through the following steps:

- (i) comparison of the performance of Quantum Transfer Learning (QTL) models with state-of-the-art classical CNNs for brain image classification,
- (ii) evaluation of computational efficiency and convergence rates of QTL models, highlighting their scalability and real-time applicability in clinical settings.
- (iii) evaluation of the robustness of QTL models in handling peripheral non-informative regions and imperfect datasets, ensuring reliable performance across diverse clinical scenarios.
- (iv) evaluation of model reliability and reproducibility through stratified five-fold cross-validation and repeated experiments using multiple random initialization seeds, along with external testing of the trained model on independent datasets to ensure stable performance under dataset shift.
- (v) incorporation of explainable artificial intelligence using Grad-CAM visualization to interpret model decisions and verify anatomically meaningful tumor localization.

III. METHODOLOGY

This section presents the proposed hybrid quantum-classical framework for multi-class brain tumor classification using

MRI. The methodology integrates anatomically guided pre-processing, deep feature extraction using the ResNet family of convolutional neural networks, and quantum-enhanced feature mapping via a variational quantum neural network (QNN). The proposed framework, termed *KCC-HQNN*, employs Kernel-based Center Cropping (KCC) as a deterministic spatial prior to emphasize anatomically relevant central brain structures by constraining the spatial field of view. Rather than acting as a learned or adaptive peripheral non-informative region removal mechanism, KCC serves to reduce peripheral non-informative regions and enforce anatomical consistency prior to feature extraction. Following classical feature extraction, a variational quantum circuit transforms the compressed feature representations into a high-dimensional Hilbert space to improve class separability. The complete workflow of the proposed framework is illustrated in Fig. 2, the integration from the linear projection layer to the quantum layer is illustrated in Fig. 3

A. DATASET SYNTHESIS AND PRE-PROCESSING

1) MULTI-SOURCE DATA INTEGRATION

Experiments were conducted on a publicly available, curated multi-class Brain MRI dataset on Kaggle, comprising 7,023 T1-weighted axial MRI images. Tumor-bearing scans corresponding to glioma, meningioma, and pituitary tumor classes were obtained from the Figshare repository, while healthy brain MRI scans representing the no-tumor category were sourced from the Br35H dataset [55]. All images were manually curated to remove duplicates and ambiguous samples, ensuring reliable ground-truth labels across four diagnostic categories. Special care was taken to mitigate potential data leakage risks common in medical imaging pipelines. All dataset splits were performed prior to preprocessing and augmentation, and no augmented samples were shared across partitions. Furthermore, images originating from the same subject were confined to a single split, preventing inadvertent information leakage that could artificially inflate performance.

2) ARTIFACT SUPPRESSION AND SPATIAL STANDARDIZATION

All MRI scans were standardized to suppress non-informative artifacts and ensure spatial consistency. Peripheral background regions, padding effects, and acquisition-induced peripheral non-informative regions were minimized. Each image was resized to a uniform spatial resolution of 224×224 pixels to ensure compatibility with the CNN model.

3) PIXEL INTENSITY NORMALIZATION

To mitigate inter-scan intensity variability, pixel values were normalized using min-max scaling:

$$I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}}, \quad (1)$$

which improves numerical stability and accelerates convergence during training.

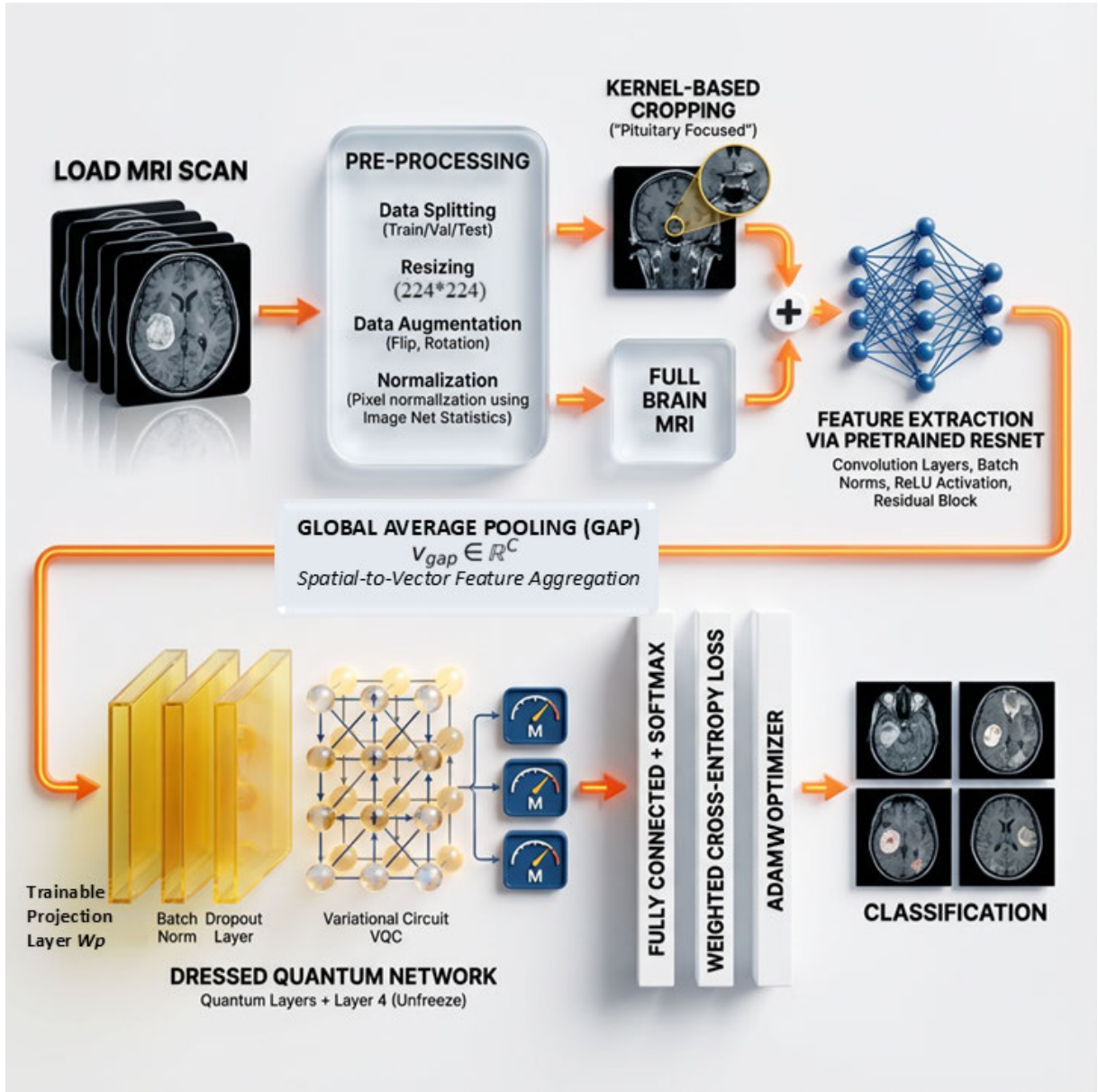


FIGURE 2. Overall workflow of the proposed KCC-HQNN model, illustrating the sequential stages from MRI data acquisition and preprocessing to classical feature extraction, quantum kernel mapping, and final multi-class classification.

4) STOCHASTIC DATA AUGMENTATION

To enhance generalization and address class imbalance, stochastic data augmentation was applied during training, including random rotations, horizontal flipping, and mild zooming. These transformations preserve anatomical realism while increasing robustness against overfitting.

B. CLASSICAL FEATURE LEARNING (CNN MODEL)

Classical feature extraction is performed using the ResNet family of convolutional neural networks, including ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152. Let $x \in \mathbb{R}^{H \times W \times C}$ denote an input MRI image. The model mapping is defined as

$$\mathbf{f} = \mathcal{M}(x; \Theta), \quad (2)$$

where $\mathcal{M}(\cdot)$ denotes the ResNet model and Θ represents trainable parameters.

Through hierarchical residual learning, early layers capture low-level intensity and edge features, while deeper layers encode semantic information related to tumor morphology, texture heterogeneity, and boundary structure. Following the final residual block, Global Average Pooling (GAP) aggregates the spatial feature maps into a high-dimensional feature vector $\mathbf{v}_{\text{gap}} \in \mathbb{R}^C$, where C depends on the chosen ResNet model (e.g., $C = 512$ for ResNet-18 and $C = 2048$ for ResNet-50).

$$\mathbf{v}_{\text{gap}} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \mathbf{f}(i, j). \quad (3)$$

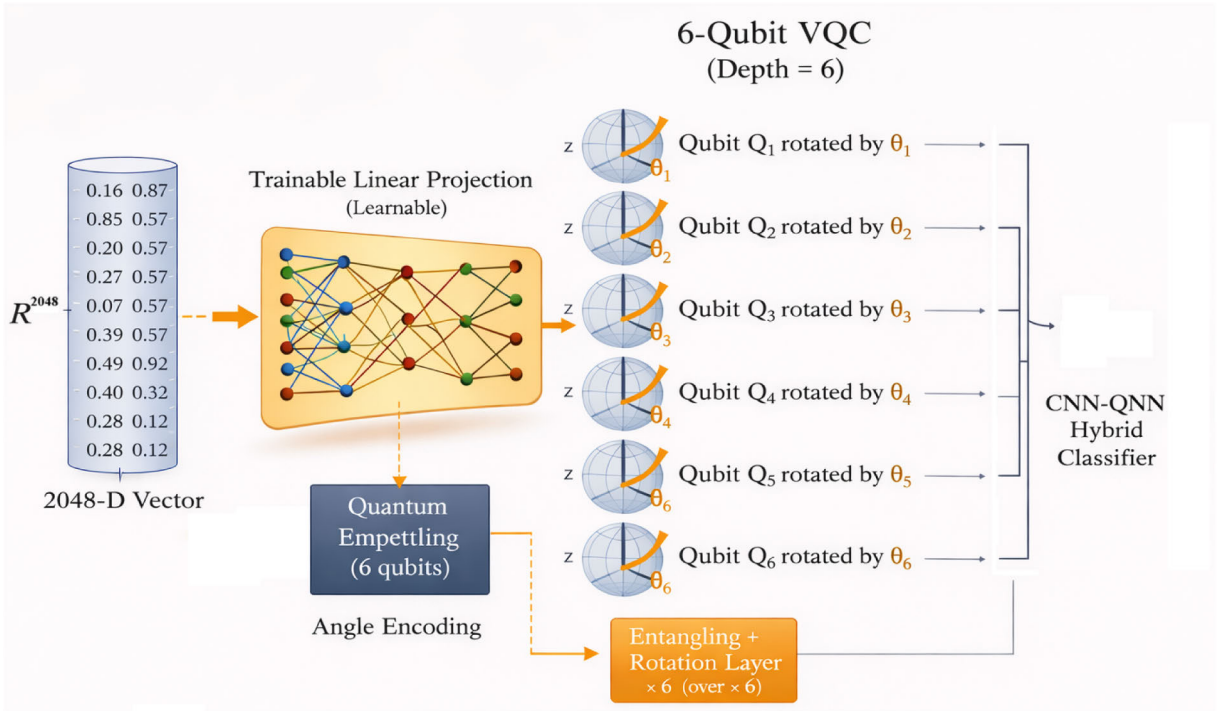


FIGURE 3. Hybrid CNN-QNN head with a 6-qubit variational quantum circuit (depth $D = 6$). The ResNet GAP feature vector ($\mathbb{R}^{2048}$) is linearly projected to $\mathbb{R}^6$, angle-encoded into six qubits, processed by entangling-rotation layers, and measured to obtain quantum features for final classification.

C. KERNEL-BASED CENTER CROPPING (KCC)

Kernel-Based Center Cropping (KCC) is a deterministic, anatomy-guided spatial preprocessing step that constrains the field of view toward clinically relevant central brain regions. The term *kernel* does not denote a learned convolution or mathematical kernel; rather, it refers to a fixed spatial cropping window applied uniformly across all samples without any trainable parameters. Each MRI slice is resized to 224×224 , and a fixed square window centered at the geometric center is used to extract the central anatomical region, which typically contains diagnostically important structures such as the pituitary gland. The cropped region is resized back to 224×224 to maintain compatibility with the ResNet model. This operation is non-adaptive and identical for all images. Unlike a standard blind center crop, KCC is motivated by neuroanatomical priors and aims to suppress peripheral non-informative regions (e.g., skull boundaries and background artifacts) while preserving central brain tissue. KCC does not guarantee noise removal, but it enforces spatial consistency prior to feature extraction. Robustness to mild spatial misalignment was verified by shifting the cropping window by ± 16 pixels during inference, resulting in only marginal performance degradation.

D. PARAMETERIZED QUANTUM FEATURE TRANSFORMATION

1) QUANTUM EMBEDDING VIA ANGLE ENCODING

Since variational quantum circuits operate on a limited number of qubits, a trainable linear projection layer is

introduced to project the high-dimensional GAP features into a six-dimensional feature vector, which is subsequently encoded into the Hilbert space of the 6-qubit quantum circuit.

The high-dimensional GAP feature vector is linearly projected via a fully connected projection layer into a six-dimensional Hilbert space:

$$\mathbf{z} = \mathbf{W}_p \mathbf{v}_{\text{gap}} + \mathbf{b}_p, \quad \mathbf{z} \in \mathbb{R}^6. \quad (4)$$

The mapping of classical features into the 2^n -dimensional Hilbert space enables the model to exploit quantum superposition, allowing it to represent complex non-linear boundaries that are computationally expensive for classical layers. This high-dimensional mapping ($\mathcal{H} = (\mathbb{C}^2)^{\otimes n_{\text{qubits}}}$) ensures that even subtle inter-class similarities in brain MRIs are separated with higher fidelity. The Bloch Sphere representation of a single qubit is shown in Fig. 4 Each component z_i is encoded into a qubit using R_Y rotations, yielding the initial quantum state

$$|\psi_{\text{in}}\rangle = \bigotimes_{i=1}^6 R_Y(z_i) |0\rangle. \quad (5)$$

2) VARIATIONAL QUANTUM CIRCUIT

The encoded state is processed by a variational quantum circuit composed of parameterized RZ – R_Y rotations and a linear CNOT entanglement chain. For circuit depth D , the

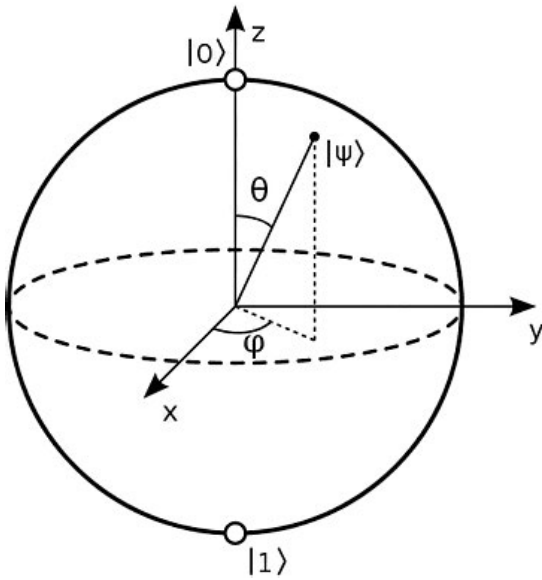


FIGURE 4. Bloch Sphere representation of a single qubit state. The mapping $R_Y(z_i)$ rotates the state vector from $|0\rangle$ towards $|1\rangle$ based on the normalized classical feature z_i , effectively encoding medical features as quantum phase information.

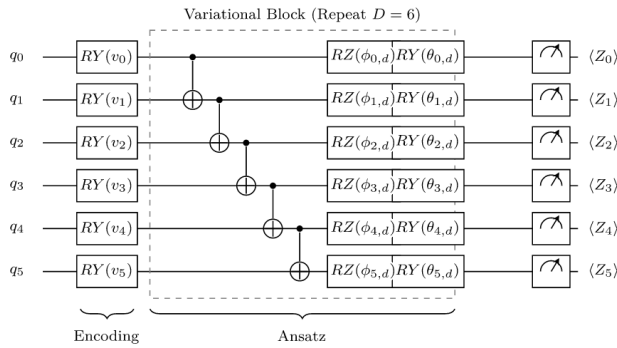


FIGURE 5. Six-qubit variational quantum circuit used for quantum kernel mapping.

output state is

$$|\psi_{\text{out}}\rangle = \prod_{d=1}^D \left[\left(\bigotimes_{j=1}^6 R_Z(\phi_{j,d}) R_Y(\theta_{j,d}) \right) \left(\prod_{i=1}^5 \text{CNOT}_{i,i+1} \right) \right] |\psi_{\text{in}}\rangle. \quad (6)$$

3) LAYER-WISE DESCRIPTION OF THE QUANTUM NEURAL NETWORK

The Quantum Neural Network (QNN) consists of a six-qubit variational quantum circuit that operates on a low-dimensional latent representation obtained from the classical CNN model. Following global average pooling, a trainable linear projection maps the high-dimensional feature vector to a six-dimensional latent space, which serves as the input to the quantum circuit.

In the first layer, each latent feature is encoded into a corresponding qubit using angle encoding via parameterized

R_Y rotations. This is followed by D variational layers, where each layer comprises parameterized single-qubit rotations (R_Z – R_Y) applied independently to all qubits, followed by a linear chain of CNOT gates to introduce entanglement across neighboring qubits, as shown in Fig. 5. The entanglement operation enables the circuit to capture non-linear correlations between feature channels that are difficult to model using classical fully connected layers.

In the final layer, expectation values of the Pauli-Z operator are measured on each qubit, producing a six-dimensional quantum feature vector. These quantum features are subsequently passed to a classical softmax classifier for multi-class prediction. All quantum circuit parameters are optimized jointly with the classical network using gradient-based training.

4) QUANTUM MEASUREMENT

Expectation values of Pauli-Z observables are measured as

$$q_i = \langle \psi_{\text{out}} | Z_i | \psi_{\text{out}} \rangle, \quad i = 1, \dots, 6. \quad (7)$$

E. HYBRID QUANTUM-CLASSICAL FUSION FRAMEWORK

The end-to-end hybrid mapping is expressed as

$$\hat{y} = \mathcal{C}(\mathcal{Q}(\mathcal{P}(\mathcal{M}(x)))), \quad (8)$$

where $\mathcal{P}$ denotes feature projection, $\mathcal{Q}$ quantum mapping, and $\mathcal{C}$ the softmax classifier.

To address class imbalance, a weighted cross-entropy loss is employed:

$$\mathcal{L}_{\text{weighted}} = - \sum_{c=1}^C w_c y_c \log(\hat{y}_c). \quad (9)$$

F. OPTIMIZATION STRATEGY AND TRAINING PROTOCOL

The hybrid model is optimized end-to-end using the AdamW optimizer. Gradients for classical parameters are computed via backpropagation, while quantum parameters are updated using the parameter-shift rule. The parameter update rule is given by

$$\Theta^{(t+1)}, \theta_q^{(t+1)} = \Theta^{(t)}, \theta_q^{(t)} - \eta \nabla \mathcal{L}. \quad (10)$$

Regularization is achieved through selective layer freezing, dropout, batch normalization, and data augmentation.

G. FIVE-FOLD CROSS-VALIDATION PROTOCOL

To obtain statistically reliable performance estimates and reduce sampling bias, we employed stratified 5-fold cross-validation. The dataset was partitioned into five mutually exclusive folds, preserving the class distribution within each fold. For each fold $k \in \{1, \dots, 5\}$, the model was trained on four folds and evaluated on the remaining held-out fold. Final performance is reported as the mean $\pm$ standard deviation across the five folds.

To prevent data leakage, fold assignment was performed prior to any preprocessing or augmentation, and no augmented samples were shared across folds. When subject-level

Algorithm 1 KCC-HQNN Hybrid Inference and Classification

Input: Brain MRI Scan (I), Pre-trained ResNet-152, QNN parameters (θ_q).

Output: Predicted Tumor Class ($C \in \{0, 1, 2, 3\}$).

- 1: **Step 1: Preprocessing**
- 2: $I_{crop} \leftarrow KCC(I)$
- 3: $I_{norm} \leftarrow \text{Normalize}(I_{crop})$
- 4: **Step 2: Feature Extraction**
- 5: $f \leftarrow \text{ResNet}(I_{norm})$
- 6: $v_{gap} \leftarrow \text{GAP}(f)$
- 7: **Step 3: Quantum Mapping (The Handoff)**
- 8: $z \leftarrow W_p \cdot v_{gap} + b_p$
- 9: $\psi \leftarrow \text{AngleEncode}(z)$
- 10: **Step 4: Quantum-Classical Fusion**
- 11: Expectation $\leftarrow \text{ExecuteQNN}(\psi, \theta_q)$
- 12: Logits $\leftarrow \text{FC}(\text{Expectation})$
- 13: **Step 5: Classification**
- 14: $C \leftarrow \text{Argmax}(\text{Softmax}(\text{Logits}))$
- 15: **return** C

identifiers were available, all slices from the same subject were kept within the same fold.

IV. RESULTS AND DISCUSSION

The proposed hybrid quantum-classical neural network has been evaluated for MR brain image classification, with results obtained across multiple ResNet architectures (ResNet-18, ResNet-34, ResNet-50, ResNet-101, and ResNet-152) integrated with a 6-qubit Quantum Neural Network (QNN).

A. DATASET AND REPRODUCIBILITY

The experiments were conducted using a publicly available 4-class Brain MRI image classification dataset curated on Kaggle, originally compiled by Nickparvar et al. [55] from the Figshare, Br35H, and Sartaj sources. The dataset consists of T1-weighted axial MRI slices. To avoid ambiguity, the evaluation protocol follows a repeated stratified holdout strategy across multiple random seeds. The exact counts of images per class and split are detailed in Table 2. To ensure the statistical reliability of the reported results and mitigate the risk of data leakage, all experiments used a stratified data partitioning strategy that prevented patient overlap across splits. Furthermore, the final metrics reported in Table 4 represent the mean performance obtained across five independent experimental runs with different random initialization seeds (seed $\in \{42, 100, 2024, 7, 555\}$). The standard deviation in accuracy across these runs was 0.2%-0.8%, validating that the proposed KCC-HQNN framework provides consistent, robust classification performance regardless of parameter initialization.

B. PERFORMANCE HIGHLIGHTS

The ResNet-18 + QNN model achieved training and validation accuracies of 99.63% and 98.76%, respectively,

TABLE 1. Summary of brain tumor classification approaches and achieved accuracies.

S. No.	Model	Dataset Used	Classification Type	Accuracy (%)	Reference
1	DeepTumorNet (CNN)	Custom MRI Dataset	3-Class	99.6	[20]
2	3ACL (Attention Conv-LSTM)	BRATS 2015, 2018	2-Class (HGG vs LGG)	98.90	[21]
3	VGG-16 + Aquila Optimizer	Custom Dataset	2-Class	97.2	[22]
4	ResNet-18 (Classical)	BraTS / Custom Dataset	4-Class	93.77	[24]
5	DenseNet (Classical)	BraTS	4-Class	94.00	[25]
6	GoogLeNet (Classical)	BraTS	4-Class	93.50	[26]
7	Xception (Classical)	Custom Dataset	4-Class	94.00	[27]
8	(Hybrid Quantum)	Custom MRI Dataset	4-Class	97.8	[28]
9	2-Qubit QNN (Quantum Hybrid)	Custom Dataset	2-Class	90.1	[29]
10	CAD System (Pituitary Tumor)	Custom Dataset	2-Class	96.00	[38]
11	Proposed KCC-HQNN (Quantum-Classical Hybrid)	Public Kaggle 4-class Brain MRI classification dataset	4-Class (Glioma, Meningioma, Pituitary, No Tumor)	99.0	Proposed Model

TABLE 2. Detailed dataset distribution per class and split.

Tumor Category	Train (70%)	Val (20%)	Test (10%)	Total
Glioma	1135	324	163	1621
Meningioma	1151	329	165	1645
No Tumor	1400	400	200	2000
Pituitary	1229	352	176	1757
Total	4915	1405	704	7023

and the rest of the models showed outstanding performance compared to the classical models ResNet-34, ResNet-50, ResNet-101, and ResNet-152, with QNN achieving 98.10%, 98.39%, 98.44%, and 99.10%, respectively. In contrast, the classical ResNet-152 model reached 95.73%, illustrating the significant performance boost a quantum neural network provides. These results validate our approach, which integrates the quantum layer into the deep learning domain of medical image analysis. The hybrid models demonstrated better generalization and enhanced robustness in distinguishing subtle differences between tumor types, as evidenced by confusion matrices. Tables 3 and 4 show the metrics for the model outcomes under the classical and quantum approaches. Training and validation curves are shown in Fig. 6. This experiment was conducted using a quantum environment simulation from the PennyLane library.

While the current framework utilizes a quantum simulator for validation, the 6-qubit Variational Quantum Circuit (VQC) is deliberately designed with limited circuit depth to minimize simulation overhead. During testing in the Kaggle

TABLE 3. Classical ResNet approach for classification of brain tumor.

Metric	ResNet-18	ResNet-34	ResNet-50	ResNet-101	ResNet-152
Learning Rate	0.001	0.001	0.001	0.001	0.001
Dropout (%)	0.05	0.05	0.05	0.05	0.05
Training Loss	0.1869	0.1775	0.1753	0.1651	0.1633
Validation Loss	0.2621	0.2289	0.2186	0.2167	0.2022
Training Accuracy (%)	94.35	94.85	94.35	94.70	95.01
Validation Accuracy (%)	93.77	93.80	93.94	93.98	94.23
Testing Loss	0.2918	0.2910	0.2893	0.2809	0.2798
Testing Accuracy (%)	92.12	92.35	92.00	92.38	93.18

TABLE 4. Quantum + ResNet approach for classification of brain tumor.

Metric	ResNet-18	ResNet-34	ResNet-50	ResNet-101	ResNet-152
Learning Rate	0.001	0.001	0.001	0.001	0.001
Dropout (%)	0.05	0.05	0.05	0.05	0.05
Training Loss	0.0283	0.0389	0.0645	0.0885	0.0558
Validation Loss	0.0536	0.0647	0.0959	0.1257	0.1469
Training Accuracy (%)	99.63	99.44	98.07	98.57	99.04
Validation Accuracy (%)	98.76	98.10	98.39	98.44	99.05
Testing Loss	0.018	0.0450	0.1220	0.11	0.12
Testing Accuracy (%)	98.25	98.00	98.20	98.22	99.00

Tesla T4 environment, the reported inference time reflects the computational cost of classical simulation of the quantum circuit on GPU hardware and does not indicate acceleration from quantum hardware execution. These results demonstrate the practical feasibility of the proposed hybrid pipeline in a simulated setting rather than real-time performance enabled by quantum processors. Any claims of real-time clinical deployment are therefore prospective and contingent upon future advances in accessible quantum processing units. Furthermore, by disentangling complex feature correlations within a high-dimensional Hilbert space, the proposed model improves the reliability of identifying subtle tumor margins, a critical requirement for clinical decision support and surgical planning.

C. TRAINING AND VALIDATION GRAPH

All models were trained for 40 epochs, and the resulting training and validation curves shown in Fig. 6 demonstrate stable convergence behavior across all ResNet models integrated with the QNN head. Both training and validation accuracy curves increase consistently while corresponding loss curves decrease smoothly, indicating effective optimization and stable learning dynamics. The validation curves closely follow the training curves with minimal divergence, demonstrating good generalization capability and limited overfitting despite the moderate size of medical imaging datasets. The incorporation of the QNN layer further stabilizes validation performance, suggesting improved feature representation compared to purely classical backbones. Among all evaluated configurations, deeper architectures

TABLE 5. Class-wise precision, recall, F1-score, support and overall metrics for ResNet + QNN models.

Model	Class	Precision	Recall	F1-Score	Support
ResNet-18 + QNN	Glioma	0.96	0.95	0.96	163
	Meningioma	0.97	0.98	0.97	165
	No Tumor	0.97	0.98	0.97	200
	Pituitary	0.98	0.98	0.98	176
	Accuracy	0.97			
ResNet-34 + QNN	Macro Average	0.97	0.97	0.97	704
	Weighted Average	0.97	0.97	0.97	704
	Glioma	0.97	0.95	0.96	163
	Meningioma	0.97	0.96	0.96	165
	No Tumor	0.97	1.00	0.98	200
ResNet-50 + QNN	Pituitary	0.99	0.98	0.99	176
	Accuracy	0.98			
	Macro Avg	0.98	0.97	0.97	704
	Weighted Avg	0.98	0.98	0.97	704
	Glioma	0.97	0.98	0.97	163
ResNet-101 + QNN	Meningioma	0.97	0.95	0.96	165
	No Tumor	0.98	0.99	0.98	200
	Pituitary	0.98	0.97	0.98	176
	Accuracy	0.98			
	Macro Avg	0.98	0.98	0.97	704
ResNet-152 + QNN	Weighted Avg	0.98	0.97	0.97	704
	Glioma	0.98	0.99	0.98	163
	Meningioma	0.98	0.99	0.98	165
	No Tumor	0.99	1.00	0.99	200
	Pituitary	1.00	0.99	0.99	176
ResNet-152 + QNN	Accuracy	0.99			
	Macro Avg	0.99	0.99	0.98	704
	Weighted Avg	0.99	0.99	0.98	704
	Glioma	0.99	0.97	0.98	163
	Meningioma	0.99	0.99	0.99	165
ResNet-152 + QNN	No Tumor	0.98	1.00	0.99	200
	Pituitary	1.00	0.99	0.99	176
	Accuracy	0.99			
	Macro Avg	0.99	0.99	0.99	704
	Weighted Avg	0.99	0.99	0.99	704

such as ResNet-101+QNN and ResNet-152+QNN achieve the highest validation performance and exhibit smoother convergence profiles, confirming their suitability for complex multiclass brain tumor classification tasks. Quantitative class-wise evaluation summarized in Table 5 shows that the proposed hybrid framework achieves an overall accuracy of up to 99%, with class-wise recall values reaching 98% for glioma, 99% for meningioma, 99% for pituitary tumors, and 100% for the no-tumor class. These results confirm consistent performance improvements across all tumor categories. The adoption of layer freezing for early residual blocks, while fine-tuning later layers and classifier heads, helps stabilize training while preventing excessive parameter overfitting. Combined with quantum feature processing, this strategy

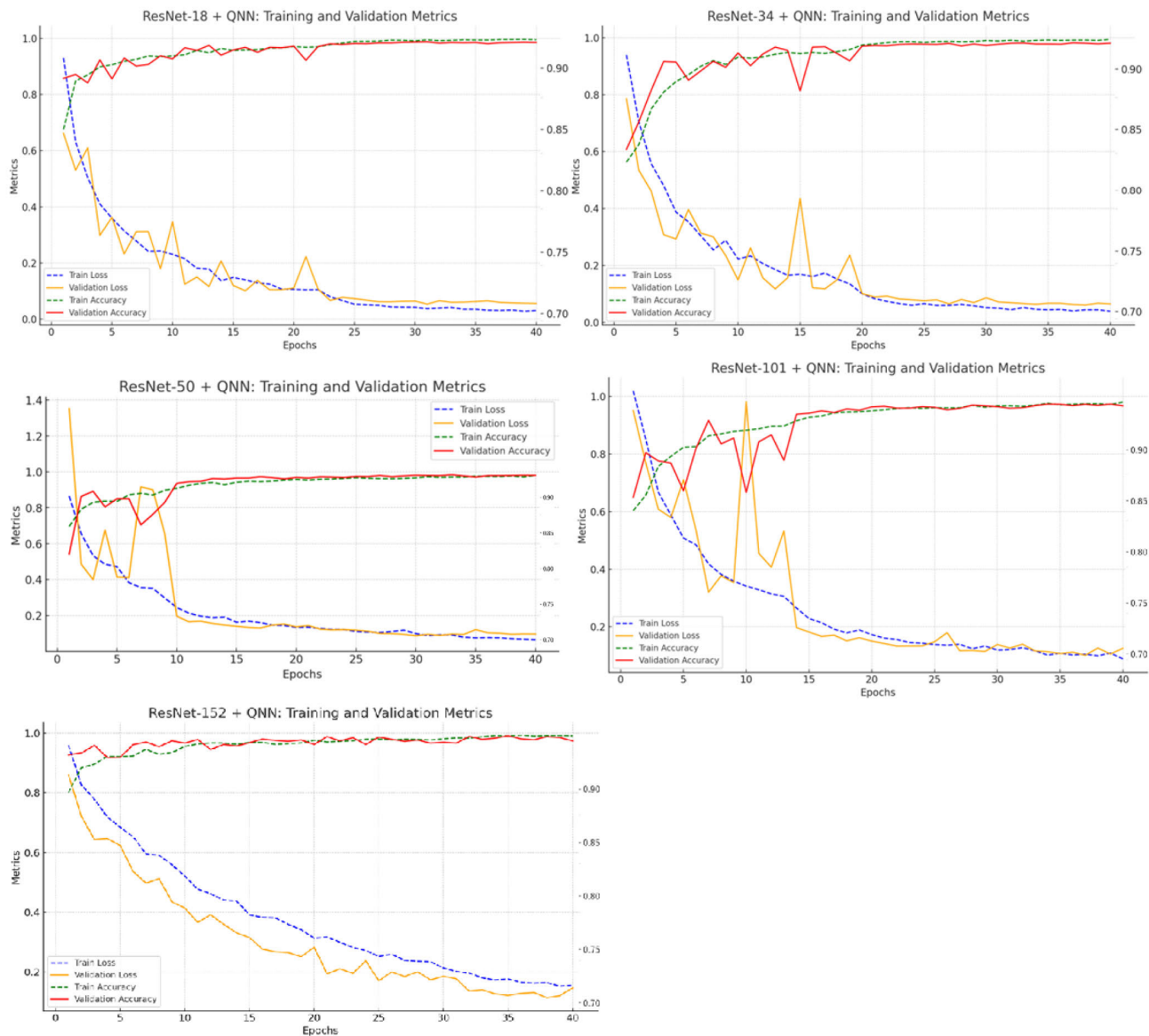


FIGURE 6. ResNet+QNN models with training and validation plot (accuracy and loss) across epochs.

enhances robustness and generalization across different tumor classes.

Overall, the training and validation trends confirm that the hybrid quantum–classical framework successfully learns discriminative representations without suffering from severe overfitting, leading to superior multiclass classification performance across all evaluated ResNet backbones.

D. IMPACT OF KCC AND QUANTUM LAYERS

This operation is particularly beneficial for detecting small, centrally located structures, such as the pituitary gland, where whole-image analysis can introduce background dominance and increase mis-classification risk, as shown in Fig. 7 and Fig. 8

To quantify the individual contributions of our proposed components, we performed a comprehensive ablation study

using the ResNet-152 model. This ensures consistency with the qualitative visualizations presented in Fig. 9. We compared baseline performance with configurations that lacked either the deterministic KCC preprocessing or the variational QNN layer in Table 6.

E. COMPUTATIONAL COMPLEXITY AND PARAMETER EFFICIENCY

To evaluate the representational power of the proposed KCC-HQNN framework, we compared the trainable parameters of the classification head. As demonstrated in Table 7, the hybrid quantum model achieves superior diagnostic accuracy while utilizing a fraction of the parameters required by a classical fully connected (FC) layer.

This drastic reduction in parameters (from thousands to just 48) highlights the efficiency of quantum feature mapping.

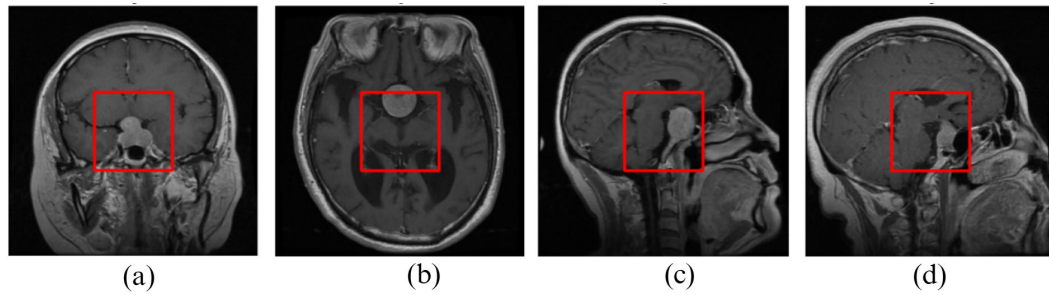


FIGURE 7. Region of interest of kernel.

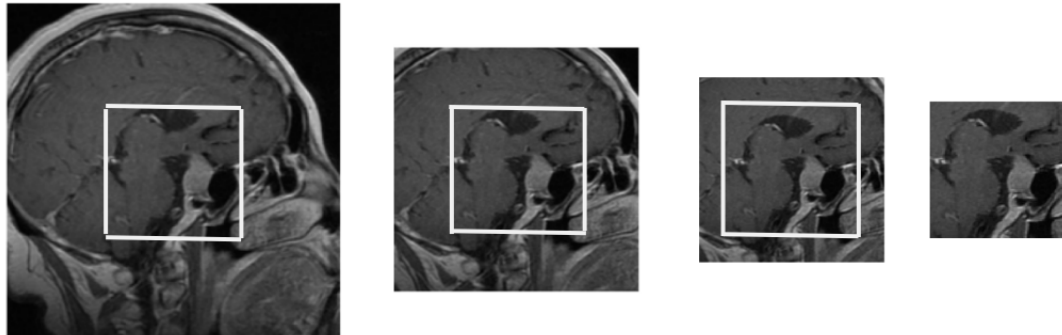


FIGURE 8. Selection of KCC.

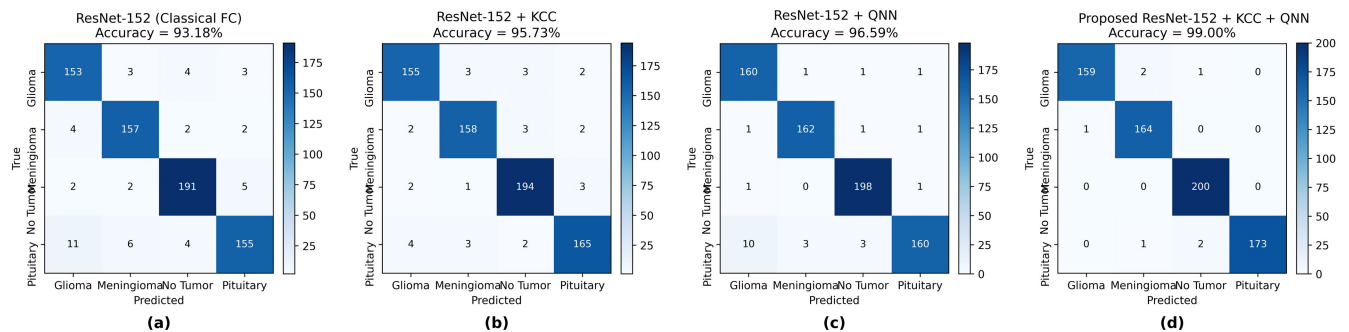


FIGURE 9. Comparison of confusion matrices for (a) classical ResNet-152, (b) ResNet-152 with KCC processing, (c) ResNet-152 with QNN head and (d) ResNet-152 with KCC and QNN head.

TABLE 6. Contribution of KCC and QNN components (ResNet-152 model).

Model	Configura-tion	Preprocessing	Classifier Head	Acc. (%)
Standard ResNet-152		Standard Resize	Classical FC	93.18
ResNet-152 + KCC (No QNN)		KCC Crop	Classical FC	95.73
ResNet-152 + QNN (No KCC)		Standard Resize	Quantum Head	96.59
KCC-HQNN (Proposed)		KCC Crop	Quantum Head	99.00

The 6-qubit Variational Quantum Circuit (VQC) exploits the 2^6 -dimensional Hilbert space to disentangle non-linear correlations between tumor features, which a classical linear layer can only achieve with much higher dimensionality and weight density.

The results in Table 6 demonstrate that while KCC significantly boosts accuracy by focusing on the anatomical

TABLE 7. Parameter efficiency of classification heads (ResNet-152 model).

Model Head	Type	Trainable Params (Head-only)	Acc. (%)
Classical Head	Linear (2048→4)	8,196	93.18
Proposed Head	6-Qubit VQC	48 (VQC angles only) [†]	99.00

[†]Excludes the lightweight classical projection and classifier layers used to interface the CNN features with the quantum circuit.

region of interest ($\sim 2.61\%$), the addition of the 6-qubit QNN layer provides a critical further enhancement ($\sim 3.27\%$). This demonstrates the contribution of the quantum variational layer in disentangling non-linear feature correlations in the high-dimensional Hilbert space. To assess the robustness of the proposed KCC-HQNN framework to spatial misalignment, we evaluated the model under off-center perturbations

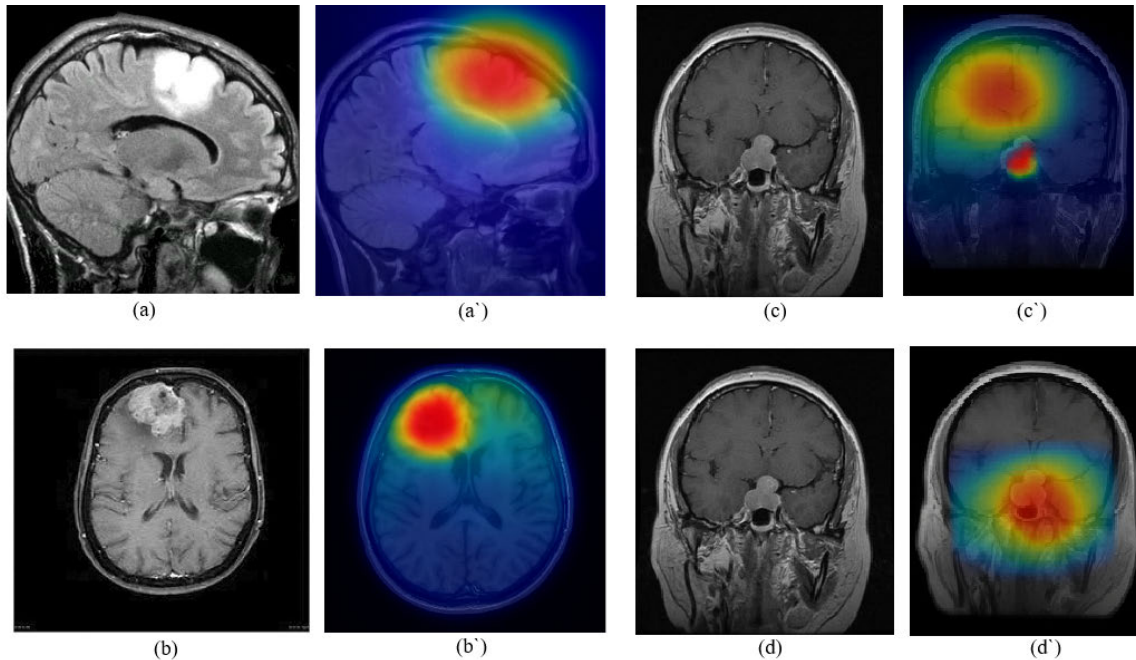


FIGURE 10. Grad-CAM visualization comparing attention maps. (a'-b') Heatmaps for glioma and meningioma cases showing precise lesion localization. (c'-d') Impact of KCC on a pituitary tumor case: where Fig. (c') without KCC and (d') attention is diffuse; with KCC, activation is tightly focused on the anatomically relevant region, effectively reducing distractions from peripheral structures. All visualizations are reported using the ResNet-152 model for consistency with the ablation analysis.

by shifting the center-cropping window by ± 16 pixels along the horizontal and vertical directions during inference. No retraining was performed. The performance under shifted crops showed only marginal degradation, indicating that the proposed framework is not overly sensitive to mild spatial misalignment and remains robust in practical settings.

F. FIVE-FOLD CROSS-VALIDATION STABILITY

Table 8 summarizes the best validation performance obtained in each fold and Table 9 presents the aggregated cross-validation statistics across folds.

TABLE 8. Five-fold cross-validation results on validation set for ResNet-152 + QNN model.

Fold	Best Epoch	Validation Accuracy (%)
1	27	99.10
2	38	98.89
3	36	99.02
4	27	98.96
5	32	99.03

G. MODEL PERFORMANCE EVALUATION ON BLIND DATASETS

To evaluate model robustness on unseen data and assess its true generalization ability, the proposed model was tested on two independent IEEE DataPort datasets, referred to as Dataset-1 and Dataset-2. Table 10 presents comparative results for ResNet models equipped with classical and quantum heads. Dataset-1 corresponds to the IEEE DataPort dataset introduced by Chaki [56], while Dataset-2 corresponds to the dataset provided by Alagur and Bhunsur-

TABLE 9. Five-fold cross-validation performance summary for ResNet-152 + QNN model.

Metric	Mean $\pm$ Std
Accuracy	99.00 $\pm$ 0.08 %
Macro F1-score	0.986 $\pm$ 0.11
Weighted F1-score	0.989 $\pm$ 0.12
Precision	0.983 $\pm$ 0.10
Recall	0.99 $\pm$ 0.12

math [57]. External testing was conducted using 10% of the available MRI samples from each dataset.

TABLE 10. Results of models' performance on two blind data sets.

Evaluation Setting	ResNet-18 (Classical)	ResNet-18 + QNN	ResNet-152 (Classical)	ResNet-152 + QNN
Dataset-1 Accuracy (%)	81.56	86.25	76.22	78.52
Dataset-2 Accuracy (%)	82.89	87.33	92.10	94.28

H. MODEL INTERPRETABILITY: GRAD-CAM VISUALIZATION

To validate the decision-making process of the proposed KCC-HQNN and ensure it identifies tumor regions rather than background artifacts, we employed Gradient-weighted Class Activation Mapping (Grad-CAM).

As illustrated in Fig. 10, the network demonstrates strong interpretability. In large lesions such as Gliomas and Meningiomas (Fig. 10a-b), the hybrid quantum-classical model accurately targets the tumor mass. More importantly, the comparison in Fig. 10(c-d) provides visual evidence for the ablation study results. The standard approach (c) exhibits scattered attention, whereas the KCC-enhanced approach (d) forces the model to prioritize the anatomically relevant

sella turcica region. This confirms that the combination of KCC and the quantum layer's high-dimensional processing acts as a deterministic spatial constraint that improves anatomical focus, thereby reducing false positives caused by peripheral non-informative regions and skull artifacts.

I. EXPERT CLINICAL VALIDATION

To assess clinical relevance, a qualitative expert review was conducted by a senior radiologist from the Department of Radiodiagnosis, IMS-BHU. The expert qualitatively examined representative classification outputs to verify anatomical plausibility and diagnostic consistency. This assessment was exploratory in nature and intended to provide domain-level insights rather than a formal clinical trial.

V. CONCLUSION

The QNN layers enhanced the model's ability to learn subtle features in the data; by including kernel-based center-cropping (KCC-HQNN), classification performance improved for small, challenging tumors, where classical models often faltered. The variational quantum circuit effectively transformed high-dimensional feature spaces, enabling more precise decision boundaries. With recall rates as high as 0.99, these models are a significant factor in missed tumor classification, which is critical for timely treatment. Despite integrating a QNN layer, the proposed model remains computationally feasible in a simulated environment and has manageable training overhead. The 6-qubit QNN configuration optimized resource usage while delivering substantial performance gains, confirming that quantum-enhanced models are theoretically viable and practically implementable on existing computational platforms.

This work provides compelling evidence of the promising potential of quantum-enhanced deep learning models for medical imaging. This approach achieves competitive state-of-the-art performance in brain tumor classification, paving the way for innovative classification tools. The practical implementation of these models in clinical workflow, advancing the field, offers new levels of precision, efficiency, and reliability in medicine. Future work will focus on benchmarking inference latency on near-term quantum processing units (QPUs) to evaluate real-time clinical applicability. This would help us gain a more qualitative insight into the model's practical strengths and weaknesses beyond simulations. We acknowledge that these experiments were conducted in a simulated quantum environment using the PennyLane library on a Kaggle cloud environment equipped with an NVIDIA Tesla T4 GPU (16 GB VRAM). The term "efficiency" refers to the expressivity of the 6-qubit Hilbert space compared to classical dense layers.

AUTHOR CONTRIBUTION

Alok Kumar Srivastava carried out the full research workflow, including study conception, dataset preparation, model design, implementation of the hybrid quantum-classical architecture, and manuscript writing. The clinical validation

of the model's outcomes was independently conducted by a senior radiologist, Dr. Ashish Verma (Professor, Department of Radiodiagnosis and Imaging, Institute of Medical Science). All authors reviewed and approved the final version of the manuscript.

DATA AVAILABILITY

The dataset used in this study is a publicly available 4-class Brain MRI image classification dataset obtained from Kaggle. The dataset contains anonymized two-dimensional MRI slices categorized into glioma, meningioma, pituitary tumor, and no-tumor classes. As the data are publicly accessible and anonymized, no ethical approval was required for this study. [55]

CONFLICT OF INTEREST

The Author declares no conflict of interest.

RESEARCH INVOLVING HUMANS AND/OR ANIMALS

This research utilizes publicly available brain tumor imaging datasets and does not involve direct experiments on human participants or animals.

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